

Power series

these are like series but you only use the ratio test or the root test when you want to find the area of convergence but when you are testing for the interval of convergence do not use the root test or the ratio test they will not work apply any other test

Refresher on root and ratio test

Root test

$$\left(\frac{n^2+1}{2n}\right)^n \Rightarrow \sqrt[n]{\left(\frac{n^2+1}{2n}\right)^n}$$

- When a whole term is raised to the power of "n" you apply root test this means you place the term in a square root to the base of "n"

$$\sqrt[n]{x^b} \Rightarrow x^{b/n}$$

$$\sqrt[n]{\left(\frac{n^2+1}{2n}\right)^n} \Rightarrow \left(\frac{n^2+1}{2n}\right)^{n/n}$$

$$\lim_{n \rightarrow \infty} \frac{n^2+1}{2n}$$

• You might need to apply L'Hospital rule

Ratio test

$$\frac{a_n}{n} \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

- a ratio test can be identified by having a "n" raised to the power and a "n" at the base

$$a_n = \frac{2^n}{n}$$

$$a_{n+1} = \frac{2^{n+1}}{(n+1)}$$

~~*~~ Simplify

$$a_{n+1} = \frac{2^n \cdot 2^1}{(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{2^n \cdot 2^1}{(n+1)} \cdot \frac{n}{2^n} \right|$$

Identifying Power series

$$\sum \frac{x^n}{(n-1)}$$

Power series can be identified by the "x" if you notice that there is one "x" while the rest are all "n" this can help you identify Power series

Finding the center

$$\sum \frac{(x-1)}{n+1}$$

to Find the center you look for the "x" and set it to zero you may also want to solve for "x" to get the center

- here highlighted is the center " $x-1=0$ "
you now just need to solve for "x"

example Problem

$$\sum \frac{(-1)^n (x+3)^n}{n \cdot 5^n}$$

$$x+3=0 \quad \text{center}$$

$$x=-3$$

$$\sum \frac{(-1)^n (x+3)^n}{n \cdot 5^n}$$

• you can factor out the $(x+3)^n$

$$(x+3) \cdot \sum \frac{(-1)^n (1)^n}{n \cdot 5^n}$$

• this has a "n" in the power and "n" in the base use ratio test

$$a_n = \frac{(-1)^n}{n \cdot 5^n}$$

$$a_{n+1} = \frac{(-1)^{n+1}}{(n+1) \cdot 5^{n+1}}$$

$$\Rightarrow \frac{(-1)^n \cdot (-1)^1}{(n+1) \cdot 5^n \cdot 5^1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(-1)^n \cdot (-1)^1}{(n+1) \cdot 5^n \cdot 5^1} \cdot \frac{n \cdot 5^n}{(-1)^n} \right|$$

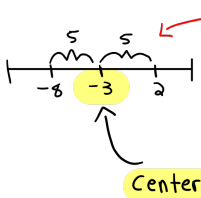
$$|x+3| \cdot \lim_{n \rightarrow \infty} \left| \frac{n}{5n+5} \right| = \frac{-\infty}{\infty} \quad \text{• apply L'Hospital rule}$$

$$L\#1 \quad (x+3) \cdot \lim_{n \rightarrow \infty} \left| \frac{-1}{5} \right| = \frac{(x+3)}{5} \quad \begin{array}{l} \bullet < 1 \text{ Converges} \\ \bullet = 1 \text{ Fails} \\ \bullet > 1 \text{ diverges} \end{array}$$

$$-1 < \frac{|x+3|}{5} < 1$$

• what ever you do on one side you should do on both sides

$$\Rightarrow \begin{array}{l} -1 < \frac{|x+3|}{5} < 1 \\ -5 < |x+3| < 5 \\ -8 < |x| < 2 \end{array}$$



the radius of convergence is 5
it must be the same on both sides

Center

$x @ -8$

$$\sum \frac{(-1)^n (x+3)^n}{n \cdot 5^n} \Rightarrow \frac{(-1)^n (-5)^n}{n \cdot 5^n}$$

$$\sum \frac{(5)^n}{n \cdot 5^n} = \sum \frac{1}{n}$$

• diverges by harmonic series
 $x @ -8$ diverges

$x @ 2$

$$\sum \frac{(-1)^n (x+3)^n}{n \cdot 5^n} \Rightarrow \sum \frac{(-1)^n (5)^n}{n \cdot 5^n}$$

$$\sum \frac{(-5)^n}{n \cdot 5^n} = \text{• apply alternating series test}$$

$$b_n = \frac{1}{n \cdot 5^n} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n \cdot 5^n} = 0 \quad \checkmark$$

$$\Rightarrow \frac{d}{dn} (n 5^n) \Rightarrow -1(n \cdot 5^n)^{-2} \cdot (5^n + n 5^n \cdot \ln(5)) \quad \checkmark$$

$$f = n \quad g = 5^n \quad \frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{df}{dn} = 1 \quad \frac{dg}{dn} = 5^n \ln(5) \quad (1)(5^n) + (n)(5^n \ln(5))$$

• Converges by the alternating series
 $x @ 2$ converges

$$\begin{array}{c} d \quad c \quad c \\ -8 \quad -3 \quad 2 \end{array} \quad \begin{array}{l} c = \text{Converges} \\ d = \text{diverges} \end{array}$$

• the center $(x=-3)$ always converges
interval of convergence

$$[-8, 2]$$

• brackets because it converges

$$\sum \frac{(-x)^n}{\sqrt[3]{n}}$$

• Center

$$-x=0 \Rightarrow \boxed{x=0}$$

$$\sqrt[n]{\frac{(-x)^n}{n^{1/3}}} \Rightarrow \frac{(-x)^{n/n}}{n^{1/3n}}$$

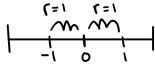
$$\lim_{n \rightarrow \infty} \frac{-x}{n^{1/3n}} = \frac{-x}{n^0} = \frac{-x}{1}$$

- < 1 Converges
- = 1 Fails
- > 1 diverges

$n^{1/3n}$ • When the denominator goes to infinite the whole thing becomes zero

$$-1 < -x < 1$$

$$-1 < x < 1$$



• radius of convergence
(= 1)

$x @ -1$

$$\frac{(-x)^n}{\sqrt[3]{n}} = \frac{(1)^n}{\sqrt[3]{n}} = \frac{1}{n^{1/3}}$$

$$p = 0.\bar{3} \quad p > 1$$

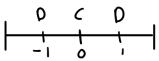
• diverges by the
p-Series test

$x @ 1$

$$\frac{(-x)^n}{\sqrt[3]{n}} \Rightarrow \frac{(1)^n}{\sqrt[3]{n}} = \frac{1}{n^{1/3}}$$

$$p = 0.\bar{3} \quad p > 1$$

• diverges by the
p-Series test



• c = converges

• d = diverges

• interval of convergence

$$\boxed{(-1, 1)}$$